

Basic Theory of Elliptic PDEs

Adaptive Finite Element Methods for Elliptic PDEs — Lecture 1

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Roadmap

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From minimization to the weak form

The Dirichlet minimization problem

- **Setup:** Let $\Omega \subset \mathbb{R}^d$ be a bounded domain, $\partial\Omega$ smooth enough to integrate by parts (C^1 or polygonal), and $f \in C(\bar{\Omega})$. Consider the **energy (Dirichlet) functional**

$$J[v] = \frac{1}{2} \int_{\Omega} |\nabla v|^2 \, dx - \int_{\Omega} f v \, dx.$$

Minimization problem

On the classical admissible set

$$\mathcal{A} = \{v \in C^1(\bar{\Omega}) : v = 0 \text{ on } \partial\Omega\},$$

$$\text{find } u \in \mathcal{A} : \quad \text{such that } J[u] = \min_{v \in \mathcal{A}} J[v].$$

- $J[v]$ is the total potential energy of a membrane clamped along $\partial\Omega$ and deflected by a transverse load f
- It is the elastic stretching energy $\frac{1}{2} \int_{\Omega} |\nabla v|^2$ minus the work $\int_{\Omega} f v$ done by the load — so its minimizer is the membrane's equilibrium shape.

Variational formulation

- **First variation:** Suppose $u \in \mathcal{A}$ minimizes J .

Fix a *test function* $v \in \mathcal{A}$, so that $u + tv \in \mathcal{A}$ for all $t \in \mathbb{R}$.

$$\begin{aligned} g(t) &= J[u + tv] = \frac{1}{2} \int_{\Omega} |\nabla(u + tv)|^2 \, dx - \int_{\Omega} f(u + tv) \, dx \\ &= J[u] + t \left(\int_{\Omega} \nabla u \cdot \nabla v - \int_{\Omega} f v \right) + \frac{t^2}{2} \int_{\Omega} |\nabla v|^2. \end{aligned}$$

- $g(0) = J[u] \leq J[u + tv] = g(t), \forall t \in \mathbb{R} \implies g(0) \leq g(t), \forall t \in \mathbb{R} \implies g'(0) = 0.$

- $g'(t) = \left(\int_{\Omega} \nabla u \cdot \nabla v - \int_{\Omega} f v \right) + t \int_{\Omega} |\nabla v|^2 \implies g'(0) = \int_{\Omega} \nabla u \cdot \nabla v - \int_{\Omega} f v$

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Weak (variational) formulation

Find $u \in \mathcal{A}$ such that

$$\mathcal{B}[u, v] := \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx =: \langle f, v \rangle \quad \forall v \in \mathcal{A}.$$

- **Equivalence:** The quadratic term $\frac{t^2}{2} \int_{\Omega} |\nabla v|^2 \geq 0$ makes J convex, so *any* $u \in \mathcal{A}$ satisfying the weak form is a minimizer. $\rightsquigarrow$ Minimization $\iff$ variational identity on $\mathcal{A}$.

Euler–Lagrange equations

$$u \in \mathcal{A} : \quad \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in \mathcal{A}.$$

- **Integrate by parts:** If $u \in C^2$, integrate the weak form by parts; the boundary term drops because $v = 0$ on $\partial\Omega$:

$$\int_{\Omega} (-\Delta u - f) v \, dx = 0 \quad \forall v \in C_0^2(\bar{\Omega}).$$

As $-\Delta u - f$ is **continuous** and this holds for all such $v \implies -\Delta u - f = 0$

Strong (Euler–Lagrange) form — the Poisson problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

- **General coefficient:**

For $J[v] = \frac{1}{2} \int_{\Omega} A \nabla v \cdot \nabla v - \int_{\Omega} f v$ with $A(x)$ spd and piecewise smooth $\implies -\operatorname{div}(A \nabla u) = f$.

Why classical spaces fail

Does a classical minimizer exist?

- **The direct method:** We showed: *if* a classical minimizer $u \in \mathcal{A}$ exists, it solves Poisson.

Q: Does one exist in $\mathcal{A}$?

Try to build it:

- J is bounded below.
- Take a *minimizing sequence* $v_n \in \mathcal{A}$ with $J[v_n] \rightarrow \inf_{\mathcal{A}} J$.
- The controlled quantity is the *energy* $\|v\|_{\mathcal{A},1} := \left(\int_{\Omega} |\nabla v|^2 \, dx \right)^{1/2}$, and (v_n) is Cauchy in it.

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The obstruction

$(\mathcal{A}, \|\cdot\|_{\mathcal{A},1})$ is **not complete**: a Cauchy sequence of smooth functions with bounded energy can converge to a function with a **foldings or mild singularities** which lies *outside* $\mathcal{A}$.

The existence tool: Riesz representation

- **Symmetric existence tool:** The clean existence tool for a *symmetric* problem:

Theorem 1 (Riesz Representation)

- Let $\mathbb{V}$ be a Hilbert space with inner product $\langle \cdot, \cdot \rangle_{\mathbb{V}}$ and norm $\|\cdot\|_{\mathbb{V}}$.
- Let $F: \mathbb{V} \rightarrow \mathbb{R}$ be a bounded linear functional ($F \in \mathbb{V}^*$).

Then there is a **unique** $u_F \in \mathbb{V}$ with

$$F(v) = \langle u_F, v \rangle_{\mathbb{V}} \quad \forall v \in \mathbb{V}, \quad \|u_F\|_{\mathbb{V}} = \|F\|_{\mathbb{V}^*} := \sup_{v \neq 0} \frac{|F(v)|}{\|v\|_{\mathbb{V}}}.$$

- **Application to the weak problem:**

- $\mathcal{B}[\cdot, \cdot]$ is **symmetric**. Is it an inner product? Check if $\mathcal{B}[v, v] = 0 \implies v = 0$.
- Is $\mathcal{A}$ complete with the norm inherited from $\mathcal{B}[\cdot, \cdot]$?
- Enlarge $\mathcal{A}$ to its *completion* in the energy norm $\rightsquigarrow$ the Sobolev space $H_0^1(\Omega)$.

Weak derivatives

Weak derivatives

- **Integration by parts:** For $u, \varphi \in C^1$ on a Lipschitz Ω (apply the Divergence Theorem to $u\varphi e_i$):

$$\int_{\Omega} u \frac{\partial \varphi}{\partial x_i} = \int_{\partial\Omega} u \varphi n_i - \int_{\Omega} \frac{\partial u}{\partial x_i} \varphi \implies \int_{\Omega} u \frac{\partial \varphi}{\partial x_i} = - \int_{\Omega} \frac{\partial u}{\partial x_i} \varphi \quad \text{if } \varphi|_{\partial\Omega} = 0.$$

- **Multi-indices:** $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d \rightsquigarrow D^{\alpha} \varphi = \frac{\partial^{\alpha_1} \dots \partial^{\alpha_d}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}} \varphi$ for any smooth φ .

- **By induction:** If $|\alpha| = \alpha_1 + \dots + \alpha_d$, $u \in C^{|\alpha|}$ and $\varphi \in C_c^{\infty}(\Omega)$,

$$\int_{\Omega} u D^{\alpha} \varphi \, dx = (-1)^{|\alpha|} \int_{\Omega} D^{\alpha} u \varphi \, dx.$$

Definition 2 (weak derivative)

Let $u \in L^1_{\text{loc}}(\Omega)$ and $\alpha \in \mathbb{N}_0^d$ a multi-index. Then $g \in L^1_{\text{loc}}(\Omega)$ is the **weak derivative** $D^{\alpha} u$ if

$$\int_{\Omega} u D^{\alpha} \varphi \, dx = (-1)^{|\alpha|} \int_{\Omega} g \varphi \, dx \quad \forall \varphi \in C_c^{\infty}(\Omega).$$

Piecewise smooth functions

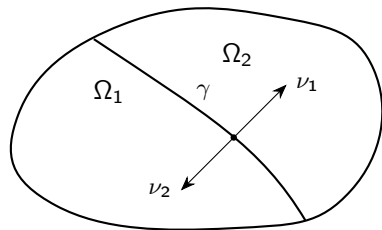
- Setup:** Let $\Omega = \Omega_1 \cup \gamma \cup \Omega_2$, $\gamma = \partial\Omega_1 \cap \partial\Omega_2$ a smooth interface, $u_i \in C^1(\overline{\Omega_i})$ with **matching traces** $u_1|_\gamma = u_2|_\gamma$, and

$$u(x) = \begin{cases} u_1(x), & x \in \Omega_1, \\ u_2(x), & x \in \Omega_2. \end{cases}$$

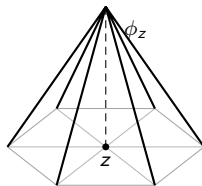
- Weak gradient:** the piecewise classical gradient:

$$\nabla u(x) = \begin{cases} \nabla u_1(x), & x \in \Omega_1, \\ \nabla u_2(x), & x \in \Omega_2. \end{cases}$$

- Application:** Continuous piecewise-linear *hat functions* ϕ_z satisfy this, so their weak gradients are the piecewise-constant $\nabla \phi_z|_T$. This is the foundation of the finite element method.



Interface γ , opposite normals $\nu_2 = -\nu_1$.



hat function ϕ_z

Piecewise smooth: proof

- **Proof:** Integrate by parts on each subdomain:

$$\begin{aligned} - \int_{\Omega} u \nabla \varphi \, dx &= - \int_{\Omega_1} u_1 \nabla \varphi \, dx - \int_{\Omega_2} u_2 \nabla \varphi \, dx \\ &= \int_{\Omega_1} \nabla u_1 \varphi \, dx + \int_{\Omega_2} \nabla u_2 \varphi \, dx - \int_{\gamma} u_1 \varphi \nu_1 \, ds - \int_{\gamma} u_2 \varphi \nu_2 \, ds. \end{aligned}$$

- **Interface cancellation:** Since $u_1 = u_2$ on γ and $\nu_2 = -\nu_1$, the two interface terms **cancel**. Hence $-\int_{\Omega} u \nabla \varphi = \int_{\Omega} (\nabla u) \varphi$ with ∇u the piecewise gradient. □

Radial functions (exercise)

Let $\rho : (0, 1) \rightarrow \mathbb{R}$ be smooth with $\int_0^1 |\rho'(r)| r^{d-1} \, dr < \infty$.

$$\text{On } \Omega = B(0, 1), \quad f(x) = \rho(|x|) \quad \implies \quad \nabla f(x) = \rho'(|x|) \frac{x}{|x|}$$

If $\rho(r) = r^\alpha$ this holds iff $\alpha > -d + 1 \rightsquigarrow \alpha$ can be negative if $d \geq 2$.

Sobolev spaces and the Sobolev number

Sobolev spaces

Definition 3 (Sobolev space)

For $1 \leq p \leq \infty$ and $m \in \mathbb{N}_0$, $W_p^m(\Omega) = \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \forall |\alpha| \leq m\}$,

$$\|u\|_{W_p^m} = \left(\sum_{|\alpha| \leq m} \|D^\alpha u\|_{L^p}^p \right)^{1/p}, \quad |u|_{W_p^m} = \left(\sum_{|\alpha|=m} \|D^\alpha u\|_{L^p}^p \right)^{1/p}, \quad (1 \leq p < \infty),$$

$$\|u\|_{W_\infty^m} = \max_{|\alpha| \leq m} \|D^\alpha u\|_{L^\infty}, \quad |u|_{W_\infty^m} = \max_{|\alpha|=m} \|D^\alpha u\|_{L^\infty}$$

- **Banach space:** $W_p^m(\Omega)$ is complete.
- **Hilbert case:** For $p = 2$, $H^m(\Omega) = W_2^m(\Omega)$ is a **Hilbert space** $\langle u, v \rangle_{H^m} = \sum_{|\alpha| \leq m} \int_\Omega D^\alpha u D^\alpha v$.
- **Zero trace:** $H_0^1(\Omega)$ denotes the closure of $C_c^\infty(\Omega)$ in $H^1(\Omega)$.

Well-posedness of Poisson in $H_0^1(\Omega)$

- **Weak formulation:** in $H_0^1(\Omega)$: find $u \in H_0^1(\Omega)$ with

$$\mathcal{B}[u, v] = \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx =: F(v) \quad \forall v \in H_0^1(\Omega).$$

- **Coercivity:** The left side is the H_0^1 inner product $\langle u, v \rangle_{H_0^1} := \mathcal{B}[u, v]$: positive definiteness rests on the zero-trace Poincaré–Friedrichs inequality (next section), $\mathcal{B}[v, v] = \|\nabla v\|_{L^2}^2 = 0 \Rightarrow v = 0$ because

$$\|v\|_{L^2(\Omega)} \leq C_{\Omega} \|\nabla v\|_{L^2(\Omega)} \quad \forall v \in H_0^1(\Omega).$$

- **Boundedness of F :** The right side is bounded:

$$|F(v)| \leq \|f\|_{L^2} \|v\|_{L^2} \leq C_{\Omega} \|f\|_{L^2} \|\nabla v\|_{L^2} = C_{\Omega} \|f\|_{L^2} \|v\|_{H_0^1} \quad \implies \quad F \in (H_0^1(\Omega))^*.$$

Well-posedness of Poisson in $H_0^1(\Omega)$

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Riesz $\Rightarrow$ well-posedness

There is a **unique** $u \in H_0^1(\Omega)$, with the stability bound $\|u\|_{H_0^1(\Omega)} = \|F\|_{(H_0^1)^*} \leq C_{\Omega} \|f\|_{L^2(\Omega)}.$

The Sobolev number

Definition

$$\text{sob}(W_p^m) = m - \frac{d}{p} \quad \rightarrow \quad \text{It appears in every scaling argument.}$$

- **Dilation:** Let $\hat{x} = \lambda x$ dilate $\Omega \rightarrow \hat{\Omega}$ and set $\hat{v}(\hat{x}) = v(x)$.

Then $D^\alpha v(x) = \lambda^{|\alpha|} D^\alpha \hat{v}(\hat{x})$, and with $d\hat{x} = \lambda^d dx$,

$$\|D^\alpha v\|_{L^p(\Omega)} = \lambda^{|\alpha| - \frac{d}{p}} \|D^\alpha \hat{v}\|_{L^p(\hat{\Omega})}.$$

- **Interpretation:** At the top order $|\alpha| = m$ the exponent is exactly

$$m - \frac{d}{p} = \text{sob}(W_p^m) \quad \rightsquigarrow \quad \text{the power of } \lambda \text{ that } |\cdot|_{W_p^m} \text{ acquires under dilation}$$

Sobolev embeddings

Motivation: radial powers $v_\gamma(x) = |x|^\gamma$

- **L^q membership:** On the unit ball $B \subset \mathbb{R}^d$, every norm is an elementary radial integral.

$$\int_B |v_\gamma|^q dx = \int_0^1 r^{\gamma q} \omega_{d-1} r^{d-1} dr = \omega_{d-1} \int_0^1 r^{\gamma q + d - 1} dr < \infty \iff \gamma q + d > 0 \iff \gamma > -\frac{d}{q},$$

$$\text{Hence, } v_\gamma = |x|^\gamma \in L^q(B) \iff \gamma > -\frac{d}{q}.$$

- **Gradient:** $\nabla v_\gamma(x) = \gamma |x|^{\gamma-1} \frac{x}{|x|}$ has $|\nabla v_\gamma| = |\gamma| |x|^{\gamma-1}$, hence

$$\nabla v_\gamma \in L^p(B) \iff (\gamma - 1)p + d > 0 \iff \gamma > 1 - \frac{d}{p}.$$

- **W_p^1 membership:**

$$v_\gamma \in W_p^1(B) \iff v_\gamma \in L^p(B) \wedge |\nabla v_\gamma| \in L^p(B) \iff \gamma > 1 - \frac{d}{p} \wedge \gamma > -\frac{d}{p} \iff \gamma > 1 - \frac{d}{p}.$$

Motivation: radial powers

- **Two thresholds:** Compare

$$v_\gamma = |x|^\gamma \in L^q(B) \iff \gamma > -\frac{d}{q}, \quad v_\gamma \in W_p^1(B) \iff \gamma > 1 - \frac{d}{p}.$$

- **Forced integrability:** If $v_\gamma \in W_p^1(B)$ then $\gamma > 1 - \frac{d}{p}$; this **forces** $v_\gamma \in L^q(B)$ exactly when

$$1 - \frac{d}{p} \geq -\frac{d}{q} \iff \frac{1}{q} \geq \frac{1}{p} - \frac{1}{d} \iff q \leq \frac{dp}{d-p} \quad (p < d).$$

This is precisely $1 - \frac{d}{p} = \text{sob}(W_p^1) \geq \text{sob}(W_q^0) = -\frac{d}{q} \quad (L^q = W_q^0).$

Sobolev embedding theorem

Theorem 4 (Sobolev embedding)

$\Omega \subset \mathbb{R}^d$: *bounded Lipschitz domain and* $m > k \geq 0$, $1 \leq p, q \leq \infty$. *If:*

- $\text{sob}(W_p^m) > \text{sob}(W_q^k)$, $(m - \frac{d}{p} > k - \frac{d}{q})$, or

- $\text{sob}(W_p^m) = \text{sob}(W_q^k)$ and $q < \infty$,

$$\implies W_p^m(\Omega) \hookrightarrow W_q^k(\Omega), \quad \text{with} \quad \|v\|_{W_q^k} \leq C \|v\|_{W_p^m}, \quad \forall v \in W_p^m(\Omega).$$

Also, the embedding is compact when the inequality is strict (Rellich-Kondrachov).

- **Remark: gaining integrability:** $m = 1, k = 0$:

$$W_p^1 \hookrightarrow L^q \quad \text{if } 1 - \frac{d}{p} \geq -\frac{d}{q} \quad \text{or} \quad q \leq \frac{dp}{d-p} \quad (\text{if } p < d).$$

- For $d = 2, p = 2$: $H^1(\Omega) \hookrightarrow L^q(\Omega)$ for all $1 \leq q < \infty$. Compact for all $1 < q < \infty$.

- For $d = 3, p = 2$: $H^1(\Omega) \hookrightarrow L^q(\Omega)$ for all $1 \leq q \leq 6$. Compact for all $1 < q < 6$.

Sobolev embedding theorem

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- **Remark: Morrey / Hölder:**

$$\text{If } \text{sob}(W_p^1) = 1 - \frac{d}{p} > 0 \quad (p > d) \quad \implies \quad W_p^1(\Omega) \hookrightarrow C^{0,\alpha}(\overline{\Omega}) \quad \text{with } \alpha = 1 - d/p.$$

The borderline failure and the role of Lipschitz

- Embedding fails when $q = \infty$ and the Sobolev numbers are equal:

If $p = d$, $\text{sob}(W_d^1) = 0 = \text{sob}(L^\infty)$. But $u(x) = \log \log \frac{2}{|x|} \in W_d^1(B_1) \setminus L^\infty(B_1)$,

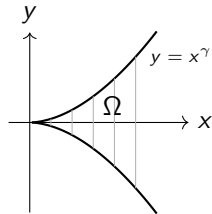
$$W_d^1(\Omega) \not\hookrightarrow L^\infty(\Omega) \quad (\text{but } W_d^1 \hookrightarrow L^q \forall q < \infty).$$

- Why Lipschitz?

On the cusp $\Omega = \{(x, y) : |y| < x^\gamma, 0 < x < 1\}$ ($\gamma > 1$)
for $p > 2$ and small $\varepsilon > 0$ with $p < \gamma + 1 - \varepsilon$

$$v(x, y) = x^{-\varepsilon/p} \in W_p^1(\Omega) \setminus L^\infty(\Omega)$$

$$\text{sob}(W_p^1) = 1 - \frac{2}{p} > 0 = \text{sob}(L^\infty) \quad \text{but} \quad W_p^1 \not\hookrightarrow L^\infty.$$



A cusp domain, $\gamma > 1$: not Lipschitz.

Useful special cases.

- $W_\infty^1(\Omega) = C^{0,1}(\overline{\Omega})$ on Lipschitz domains.
- For nodal interpolation ($d = 2$): $H^2 \hookrightarrow C^0$ ($\text{sob} = 1$),
- The borderline $W_1^2 \hookrightarrow C^0$ ($\text{sob} = 0$) as well.

Density

Density of smooth functions

- **Why density:** Smooth functions approximate Sobolev functions — this is what lets us prove estimates on smooth representatives and pass to the limit.

Theorem 5 (density, Lipschitz domains)

*If $\Omega \subset \mathbb{R}^d$ is Lipschitz and $1 \leq p < \infty$, then $C^\infty(\overline{\Omega})$ is dense in $W_p^m(\Omega)$:
for every $u \in W_p^m(\Omega)$ there exists $(u_n) \subset C^\infty(\overline{\Omega})$ with $\|u - u_n\|_{W_p^m(\Omega)} \rightarrow 0$.*

Theorem 6 (density, open domains)

If $\Omega \subset \mathbb{R}^d$ is open and $1 \leq p < \infty$, then $C^\infty(\Omega) \cap W_p^m(\Omega)$ is dense in $W_p^m(\Omega)$.

Failure at $p = \infty$

Both fail for $p = \infty$: a uniform limit of continuous functions is continuous, but $C^0(\overline{\Omega}) \subsetneq L^\infty(\Omega)$.

Poincaré–Friedrichs inequalities

Two Poincaré–Friedrichs inequalities

• One idea, two forms:

If “constant functions are excluded” the full W_p^1 norm is controlled by the gradient alone.

Zero trace — drives coercivity (Section 1)

For Ω bounded, $1 \leq p < \infty$, there is C_Ω depending only on $\text{diam } \Omega$ and p with

$$\|v\|_{L^p(\Omega)} \leq C_\Omega \|\nabla v\|_{L^p(\Omega)} \quad \forall v \in W_{p,0}^1(\Omega) = \overline{C_c^\infty(\Omega)}^{W_p^1} \quad (\text{so } W_{2,0}^1 = H_0^1).$$

Hence $|v|_{W_{p,0}^1(\Omega)} = \|\nabla v\|_{L^p(\Omega)} =: \|v\|_{W_{p,0}^1(\Omega)}$ is a norm, equivalent to $\|\cdot\|_{W_p^1}$ in $W_{p,0}^1(\Omega)$.

Zero mean — drives interpolation (Section 2)

For B bounded, connected, Lipschitz and $1 \leq p \leq \infty$, there is C_P with

$$\|g\|_{L^p(B)} \leq C_P |g|_{W_p^1(B)} \quad \text{for all } g \in W_p^1(B), \int_B g \, dx = 0.$$

Zero-trace Poincaré: proof (constant = diam Ω)

Lemma 1

$\|v\|_{L^p(\Omega)} \leq C_\Omega \|\nabla v\|_{L^p(\Omega)}$ for $v \in W_{p,0}^1(\Omega)$, with $C_\Omega = \text{diam } \Omega$.

Proof.

- **Reduce to smooth v :** By density it suffices to treat $v \in C_c^\infty(\Omega)$. Translate so $\Omega \subset \{0 < x_1 < L\}$, $L = \text{diam } \Omega$. Extend v by zero and use the fundamental theorem of calculus in x_1 :

$$v(x) = \int_0^{x_1} \partial_1 v(t, x') dt, \quad x = (x_1, x').$$

- **Hölder in x_1 :** For $p > 1$,

$$|v(x)|^p \leq x_1^{p-1} \int_0^L |\partial_1 v(t, x')|^p dt \leq L^{p-1} \int_0^L |\partial_1 v(t, x')|^p dt$$

(for $p = 1$ the factor x_1^{p-1} is absent). Integrate in $x_1 \in (0, L)$, then in x' :

$$\|v\|_{L^p(\Omega)}^p \leq L^p \int_\Omega |\partial_1 v|^p dx \leq L^p \|\nabla v\|_{L^p(\Omega)}^p,$$

i.e. $C_\Omega = \text{diam } \Omega$.

- **Norm equivalence:** follows from $\|\nabla v\|_{L^p} \leq \|v\|_{W_p^1} \leq (1 + C_\Omega) \|\nabla v\|_{L^p}$. □

The energy inner product

Why the lemma matters

The zero-trace Poincaré inequality is exactly what makes the Dirichlet form $\mathcal{B}[u, v] = \int_{\Omega} \nabla u \cdot \nabla v \, dx$ an **inner product** on $H_0^1(\Omega)$:

- bilinearity and symmetry are immediate;
- positive definiteness is $\mathcal{B}[v, v] = \|\nabla v\|_{L^2}^2 = 0 \Rightarrow v = 0$, which is the case $\|v\|_{L^2} \leq C_{\Omega} \cdot 0$ of the lemma.

- **Consequence:** This is the fact invoked (by forward reference) in the well-posedness argument: the same bound $\|v\|_{L^2} \leq C_{\Omega} \|\nabla v\|_{L^2}$ shows the load functional F is bounded, closing the Riesz argument for the Poisson problem.

Zero-mean Poincaré: proof (compactness)

Lemma 2

For B bounded, connected, Lipschitz, $1 \leq p \leq \infty$: every $g \in W_p^1(B)$ with $\int_B g \, dx = 0$ obeys $\|g\|_{L^p(B)} \leq C_P \|\nabla g\|_{L^p(B)}$.

Proof (by contradiction).

- **Assume failure:** If no C_P worked, there would be $g_n \in W_p^1(B)$ with

$$\int_B g_n \, dx = 0, \quad \|g_n\|_{L^p(B)} = 1, \quad |g_n|_{W_p^1(B)} \rightarrow 0.$$

- **Compactness:** The g_n are bounded in $W_p^1(B)$, so by **Rellich–Kondrachov** (compactness of $W_p^1(B) \hookrightarrow L^p(B)$ on Lipschitz B) a subsequence converges in $L^p(B)$ to some g . Since $\nabla g_n \rightarrow 0$ in L^p too, the sequence is Cauchy in $W_p^1(B)$ and the limit has $\nabla g = 0$; hence g is **constant** on the connected B .
- **Contradiction:** Passing to the limit in $\int_B g_n = 0$ gives $\int_B g = 0$, so the constant is 0, i.e. $g \equiv 0$ — contradicting $\|g\|_{L^p(B)} = \lim \|g_n\|_{L^p(B)} = 1$. □

Two faces of one idea

Summary of the pair

Each lemma removes the kernel of the gradient by a *normalization*:

- a **boundary condition** (zero trace) — drives coercivity and well-posedness (Section 1);
- a **zero-mean condition** — drives polynomial approximation and interpolation estimates (Section 2).

- **One proof, two lemmas:** The contradiction proof of the zero-mean lemma applies *verbatim* to the zero-trace lemma (replace the mean constraint by the trace constraint). We gave the direct argument for the zero-trace case because it also yields the **explicit** constant $C_\Omega = \text{diam } \Omega$.

Nonsymmetric problems and Lax–Milgram

Beyond symmetry

- **Symmetry exploited:** Riesz representation exploits **symmetry**: $\mathcal{B}[\cdot, \cdot]$ is itself an inner product, and u is the Riesz representative of F .
- **Nonsymmetric case:** For a **nonsymmetric** problem — a convection term $\mathbf{b} \cdot \nabla u$ — the form is no longer symmetric and this argument breaks down. The replacement:

Theorem 7 (Lax–Milgram)

- $\mathbb{V}$: *Hilbert space*
- $\mathcal{B}[\cdot, \cdot] : \mathbb{V} \times \mathbb{V} \rightarrow \mathbb{R}$ *bilinear with*

$$|\mathcal{B}[u, v]| \leq M \|u\|_{\mathbb{V}} \|v\|_{\mathbb{V}} \text{ (continuity)}, \quad \mathcal{B}[v, v] \geq \alpha \|v\|_{\mathbb{V}}^2 \text{ (coercivity)}, \quad (M, \alpha > 0)$$

- $F : \mathbb{V} \rightarrow \mathbb{R}$ *linear and bounded*: $|F(v)| \leq \Lambda \|v\|_{\mathbb{V}}$

Then there is a **unique** $u \in \mathbb{V}$ with $\mathcal{B}[u, v] = F(v) \quad \forall v \in \mathbb{V}$, and $\|u\|_{\mathbb{V}} \leq \Lambda/\alpha$.

Lax–Milgram: proof sketch

Proof (sketch).

- **The operator A :** Fix $u \in \mathbb{V}$. Since $v \mapsto \mathcal{B}[u, v]$ is a bounded linear functional, Riesz gives a unique $Au \in \mathbb{V}$ with $\mathcal{B}[u, v] = \langle Au, v \rangle_{\mathbb{V}}$ for all v . The map $A : \mathbb{V} \rightarrow \mathbb{V}$ is linear and bounded ($\|Au\|_{\mathbb{V}} \leq M \|u\|_{\mathbb{V}}$). The equation $\mathcal{B}[u, v] = F(v) = \langle u_F, v \rangle$ becomes $Au = u_F$.
- **Bounded below:** Coercivity gives

$$\alpha \|u\|_{\mathbb{V}}^2 \leq \mathcal{B}[u, u] = \langle Au, u \rangle_{\mathbb{V}} \leq \|Au\|_{\mathbb{V}} \|u\|_{\mathbb{V}} \Rightarrow \|Au\|_{\mathbb{V}} \geq \alpha \|u\|_{\mathbb{V}},$$

so A is bounded below — injective with closed range.

- **Surjectivity:** follows from a Banach fixed-point iteration $u \leftarrow u - \rho(Au - u_F)$ for small $\rho > 0$, giving existence and uniqueness. □

Riesz is the symmetric special case

When $\mathcal{B}[\cdot, \cdot]$ is symmetric it defines an equivalent inner product and A is the identity (up to scaling), so Lax–Milgram reduces to Riesz. For the model Poisson problem, Riesz already suffices.

A general elliptic model problem

Nonsymmetric second-order elliptic problem

$$\begin{cases} -\operatorname{div}(A\nabla u) + \mathbf{b} \cdot \nabla u + cu = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

$A \in L^\infty(\Omega; \mathbb{R}^{d \times d})$ symmetric uniformly positive definite, $\mathbf{b} \in L^\infty(\Omega; \mathbb{R}^d)$, $c \in L^\infty(\Omega)$, $f \in L^2(\Omega)$.

- **Bilinear form:** Set $\mathbb{V} = H_0^1(\Omega)$, $\|v\|_{\mathbb{V}} = \|\nabla v\|_{L^2}$. The bilinear form

$$\mathcal{B}[u, v] = \int_{\Omega} A \nabla u \cdot \nabla v + \int_{\Omega} (\mathbf{b} \cdot \nabla u) v + \int_{\Omega} cu v.$$

- **Continuity:** immediate from Hölder and the L^∞ bounds.

Coercivity and conclusion

- **Coercivity:** uses $A(x)\xi \cdot \xi \geq \lambda |\xi|^2$ a.e. and the antisymmetry of the convection term. Integration by parts (with $v = 0$ on $\partial\Omega$):

$$\int_{\Omega} (\mathbf{b} \cdot \nabla v) v \, dx = -\frac{1}{2} \int_{\Omega} (\operatorname{div} \mathbf{b}) v^2 \, dx.$$

- **Conclusion:** Therefore

$$\begin{aligned} \mathcal{B}[v, v] &= \int_{\Omega} A \nabla v \cdot \nabla v \, dx + \int_{\Omega} \left(c - \frac{1}{2} \operatorname{div} \mathbf{b}\right) v^2 \, dx \\ &\geq \lambda \|\nabla v\|_{L^2}^2 + \int_{\Omega} \left(c - \frac{1}{2} \operatorname{div} \mathbf{b}\right) v^2 \, dx \\ &\geq \lambda \|\nabla v\|_{L^2}^2, \end{aligned}$$

provided $c - \frac{1}{2} \operatorname{div} \mathbf{b} \geq 0$ a.e.

Hence **Lax–Milgram applies**: a unique $u \in H_0^1(\Omega)$ solves the problem weakly.

Summary

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- ① **Minimize** $\rightarrow$ **weak** $\rightarrow$ **strong**: $J[v] = \frac{1}{2} \int |\nabla v|^2 - \int f v$ leads to $\mathcal{B}[u, v] = \langle f, v \rangle$, whose Euler–Lagrange form is $-\Delta u = f$.
- ② **Classical spaces fail**: the energy norm is not complete on $\mathcal{A}$; weak solutions may be singular (log-log example, fundamental solution).
- ③ **Weak derivatives & Sobolev spaces**: $W_p^m(\Omega)$, Banach; H^m Hilbert. The **Sobolev number** $\text{sob}(W_p^m) = m - d/p$ governs all scaling.
- ④ **Embeddings**: $W_p^m \hookrightarrow W_q^k$ if $\text{sob}(W_p^m) \geq \text{sob}(W_q^k)$ (except when $q = \infty$, in which case inequality must be strict.)
- ⑤ **Poincaré–Friedrichs**: $\|v\|_{L^p} \leq \text{diam } \Omega \|\nabla v\|_{L^p}$ (zero trace) $\Rightarrow$ energy is a norm on H_0^1 ; zero-mean version drives interpolation.
- ⑥ **Existence**: Riesz (symmetric) and Lax–Milgram (nonsymmetric) give unique weak solutions with stability $\|u\|_{\mathbb{V}} \leq \Lambda/\alpha$.

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Next: Lecture 2 — finite element discretization and interpolation theory.